

Practical -11

1) Race

```
bharath_sai_reddy@LAPTOP-1  ×  +  v
GNU nano 7.2 race.c *
#include <stdio.h>
#include <pthread.h>

int counter = 0;

void *increment(void *arg)
{
    for (int i = 0; i < 100000; i++)
    {
        counter++;
    }

    return NULL;
}

int main()
{
    pthread_t t1, t2;

    pthread_create(&t1, NULL, increment, NULL);
    pthread_create(&t2, NULL, increment, NULL);

    pthread_join(t1, NULL);
    pthread_join(t2, NULL);

    printf("Final counter: %d\n", counter);

    return 0;
}
```

```

bharath_sai_reddy@LAPTOP-1  ×  +  ▾
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/practical-12$ nano deadlock_prevention.c
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/practical-12$ gcc deadlock_prevention.c -o de
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/practical-12$ ./deadlock_prevention
Thread 1: locking Resource 1
Thread 2: locking Resource 1
Thread 1: locking Resource 2
Thread 1: acquired both resources
Thread 2: locking Resource 2
Thread 2: acquired both resources
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/practical-12$ nano deadlock_prevention.c
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/practical-12$ nano deadlock.c
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/practical-12$ nano producer_consumer.c
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/practical-12$ cd ..
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical$ cd practical-11
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/practical-11$ nano race.c

bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/practical-11$ gcc race.c -o race -pthread
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/practical-11$ ./race
Final counter: 128622
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/practical-11$ ./race
Final counter: 100000
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/practical-11$ ./race
Final counter: 144317
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/practical-11$ ./race
Final counter: 105458
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/practical-11$ ./race
Final counter: 103983
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/practical-11$ ./race
Final counter: 112329
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/practical-11$ ./race
Final counter: 100000
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/practical-11$ ./race
Final counter: 200000
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/practical-11$ ./race
Final counter: 128647
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/practical-11$ |

```

2) Mutex

```

bharath_sai_reddy@LAPTOP-1  ×  +  ▾
GNU nano 7.2                                     mutex.c *
#include <stdio.h>
#include <pthread.h>
int counter = 0;
pthread_mutex_t lock;
void *increment(void *arg)
{
    for (int i = 0; i < 100000; i++)
    {
        pthread_mutex_lock(&lock);
        counter++;
        pthread_mutex_unlock(&lock);
    }
    return NULL;
}
int main()
{
    pthread_t t1, t2;
    pthread_mutex_init(&lock, NULL);
    pthread_create(&t1, NULL, increment, NULL);
    pthread_create(&t2, NULL, increment, NULL);
    pthread_join(t1, NULL);
    pthread_join(t2, NULL);
    printf("Final counter: %d\n", counter);
    pthread_mutex_destroy(&lock);
    return 0;
}

bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/practical-11$ ./mutex
Final counter: 200000

```